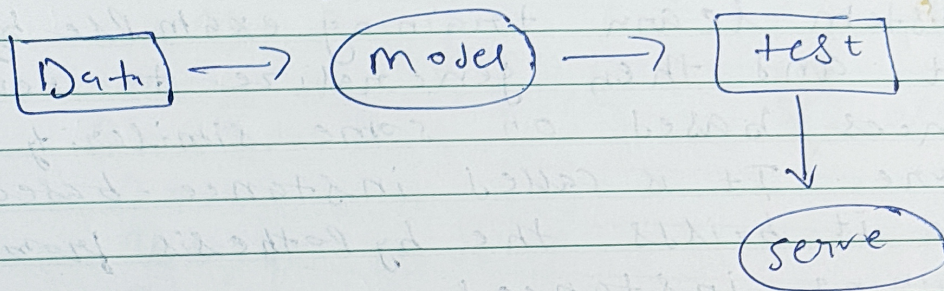


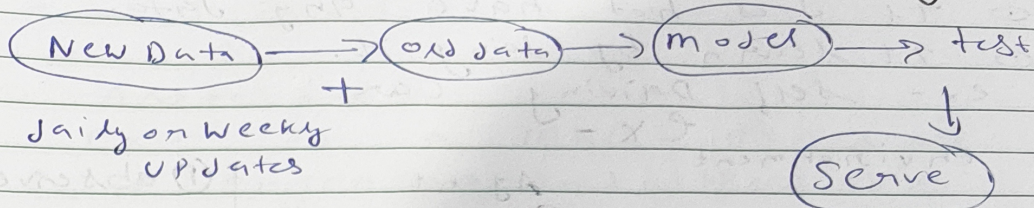
found

Batch / offline ML
the system collect data over a
period of time and process it all at once
in a large group or batch.





Problem With Batch learning



- ⊙ Lot of Data
- ⊙ Hardware Limitation
- ⊙ Availability

⊙ Online ML

- also called incremental learning
- model constantly updates as new data arrives. It doesn't wait for batch, it learn from data point one by one.

When to use?

- ⊙ constant changing environment like stock market.
- ⊙ news.
- ⊙ Cost Effective
- ⊙ Faster solution.

Instance Based Learning

System learn training example by heart and then generalize to new instances based on some similarity measure. It is called instance-based because it builds the hypothesis from training instances.

It is also known as memory-based learning.



- ① Time complexity of this algorithm depends upon size of training data.
Worst-case time complexity of this algo. is $O(n)$ where n is number of training instances.

Model-based learning

A system is called model-based when it learns from the data and creates a model which has some parameters and it predicts the output by using this training data.

Challenges in ML

1. Data collection
2. Insufficient Data / Labelled Data
3. Non-Representative Data
4. Poor Quality Data
5. Irrelevant Feature / Column
6. Overfitting
7. Underfitting
8. Software Integration
9. Offline learning / Deployment
10. Cost Involved

ML D L C

- ① Framing the Problem
- ② Data Preparation
- ③ Data Wrangling
- ④ Analyse Data
- ⑤ Train model
- ⑥ Test model
- ⑦ Deploy model



Frame Problem
Gathering Data
Data Preprocessing
EDA
Feature Engineering
Model Training, Evaluation
Model Deployment
Testing
Optimize